

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	7.145158 d	VSX	no match
Epoch (t0)	1546.7268 BTJD	ASAS-SN	no match
Duration	0.0840 d	Gaia eclipsing-binary table	no match
Depth	5.137 % (51,365 ppm)	Gaia variable classification	no match
SNR	209.2	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	4 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.06 (gate >= 0.50)		
Robust depth sigma	86.0 (gate >= 7.0)		
Odd/even z-score	0.51 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

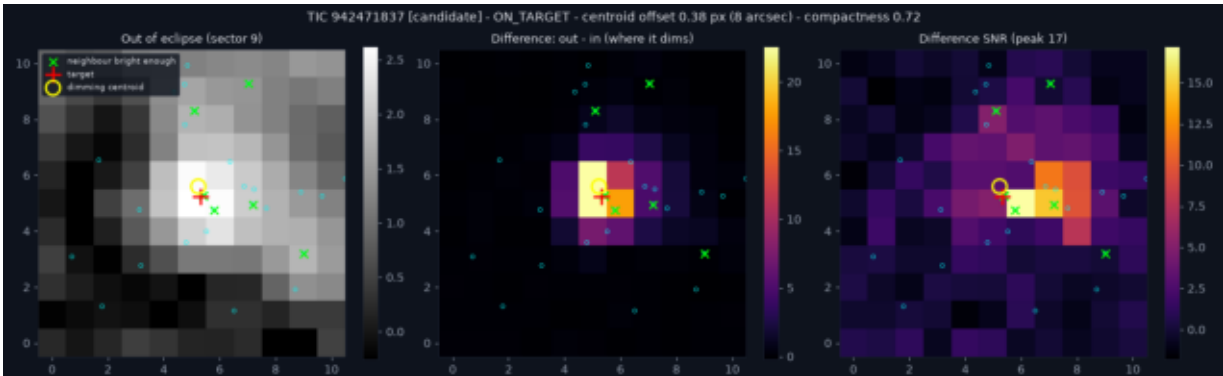
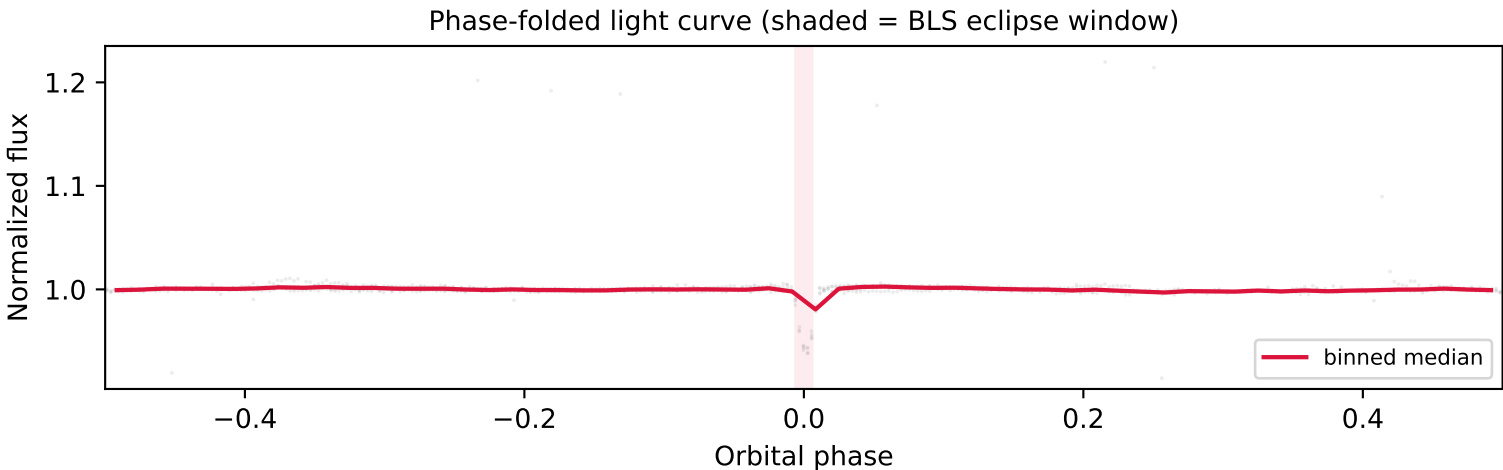
Sector 9 light-curve dip in this sector: 74.3 sigma t0 refined by 0.0000 phase

Difference-image peak SNR 17.18 (gate >= 5.0) Compactness 0.72 (gate >= 0.50)

Dimming centroid offset 0.38 px (8 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 23382413

0.38 px away, dTmag -0.65 - farther from the dimming than the target



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.002 px (dimming centroid 0.376 px from target vs 0.378 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.